OMIS 30 – Introduction to Programming

Spring 2024, Apr 2 – Jun 7

Class Time: Section-1 MWF 9:15am – 10:20am

Section-2 MWF 10:30am – 11:35am

Class Location: Lucas Hall 309

Professor Yaqiong (Joan) Wang

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Office Hours: Wednesday 4:30pm - 5:30pm or by appointment

Office Location: Lucas Hall 216D

COURSE OVERVIEW

This course is an introduction to the fundamental principles of programming for students without any programming background. A set of core programming basics – including data types, control structures, algorithms development, and program design with functions – will be introduced using Python as the programming language throughout this course. As a language with a simple syntax and a powerful set of libraries, Python is easy for beginners to learn. At the same time as an interpreted language, with a rich programming environment including a robust debugger and profiler, Python is also widely used in many scientific areas for data analysis. Students should have their laptops / PCs and internet access to use the Python Integrated Development Environment during the sessions. Through a combination of readings, lectures, examples, labs, and assignments, we will learn the following topics:

- Python setups (Installation, Text Editors, Interactive shell, Editing and running a script etc.)
- Algorithms and Information Processing
- Boolean logic and Numeric Data Types
- Strings, Text Files, Lists, and Dictionaries
- Control structures
- Functions and Classes
- Packages and Libraries
- Using python for basic data exploration and analysis

At the end of this course, students are expected to master Python fundamentals and be able to:

- Understand and apply fundamental programming principles, coding practices and tools
- Read and interpret most Python code
- Identify/characterize/define a problem
- Design a program to solve a problem
- Create executable code to solve a problem
- Run and test code to solve a problem
- Create simple and practical applications

INSTRUCTOR INFORMATION

I joined Leavey School of Business as an Assistant Professor in Fall 2020 after I earned my PhD from University of Minnesota. My research interests include recommender systems, multichannel retailing, machine learning

application and evaluation. My programming experience dates back to 2009 when I was a sophomore in college. I learned both C and Java during my undergrad studies and was involved in a real-world system development project which uses Java as the primary programming language. I started using Python for my research projects from 2012 and it has been my favorite and mostly frequently used programming language for data analysis since then. I believe my personal experiences in learning and using programming languages can bring extra references and value to the classroom.

I love teaching and believe in active learning. In my class, in addition to transmitting knowledge, I will also try various ways to create a free and stimulating environment for my students to take charge and independently construct their own knowledge. If you have any questions or need any help, please feel free to reach out to me. Email is the best way to reach me, and I typically respond within 24 hours.

COURSE MATERIALS

Required Text: Programming in Python 3 (Online zyBook, ISBN: 979-8-203-34387-1).

zyBook subscription instruction:

- Sign in or create an account using your scu email at learn.zybooks.com
- Enter zyBook code: SCUOMIS30WangSpring2024
- Subscribe

A subscription fee is $89. Students may begin subscribing on Apr 1, 2024 and the cutoff to subscribe is Jun 30, 2024. Subscription will last until Jun 30, 2023.

Students can either subscribe directly on the zyBook website and pay with a credit or debit or can purchase an access code through the SCU bookstore and type in the access code on the zyBook website.

Students typically subscribe directly from zyBook website, but those that use financial aid or scholarship should go through the bookstore.

Supplementary Text: Mark Lutz, Learning Python, 4th Edition, O'Reilly.

Additional Materials: Lecture slides, updates, etc. will be posted on our Camino site:

Section 1/2: https://camino.instructure.com/courses/97727

Note that some lectures might include materials not covered in the textbook. Check our Camino site regularly for updates.

You are responsible to ensure that your email address listed on Camino is correct and that you do not miss important announcements or other information. "Not knowing" is not an acceptable excuse.

Other Useful Resources: https://docs.python.org/3/

https://www.stackoverflow.com

https://www.learnpython.org

https://learnpythonthehardway.org/book/appendixa.html

http://www.pythontutor.com/visualize.html#mode=edit

https://openai.com/blog/chatgpt

COURSE REQUIREMENTS

1. In-Class Coding Practices / Attendance (10 points)

10% of the grade is based on in-class coding practices. In-class coding practices will involve solving problems that are similar to the examples I demonstrate during each session, both independently and with your group.

No submission is needed for the in-class coding practices, so this portion of grade will be mainly based on my observation or random check of student engagement during the session. For example, I might randomly call on some group to share the problems they encounter during the coding and how the problems are eventually resolved, I might walk around the classroom to check on students' coding progress, or I might require some students (2 for each practice at the end of the session) to turn in their coding practice.

Class attendance is obviously required to get full credits in this case.

Attendance exceptions will be granted in rare circumstances only with prior approval. Please let me know as much in advance as possible if there is a need to miss a class.

2. Knowledge Checks (10 points)

10% of the grade is based on knowledge checks which include *reading contents* and *completing participation activity* in the online textbook.

Readings related to our lectures and in-class coding practices will be assigned on Camino site about one week before each session (except for the first session). Students are recommended to finish the assigned readings before the lecture session. To get full credits, students are expected to complete the participation activity as knowledge checks after the lecture session. Your activity report will be automatically generated from zyBooks system and be used to calculate this portion of your grade.

The purpose of the pre-class reading is to encourage student engagement and make sure I can focus on key concepts, demonstrate examples and allow more time for in-class coding practices or student group work, which is often critical for effective learning of a programming language.

3. Examinations (40 points)

40% of the grade is based on exams which include 2 Quizzes (5 points each), a Midterm exam (10 points) and a Final exam (20 points).

All exams will be open book / Internet (except for ChatGPT). Each will be comprehensive and cover all materials assigned up to the date of the exam.

Makeup tests will not be given unless for serious legitimate excuses allowed by university policies and arranged in advance.

4. Homework (40 points)

40% of the grade is based on homework. There will be 8 homework assignments throughout the course (5 points each). One each week after the first week, except for the last week.

Solutions for homework problems that asking for writing a python program must include full working code to get full credits.

More details (due dates, deliverables, grading) about the homework will be elaborated in each assignment.

GRADING POLICY

Your final grade in this course will be based on the following.

	Points	Comments
Knowledge Checks	**10**	
Readings	-	For around 10 sections
Participation Activity	10	For around 10 assignments
In-Class Coding Practices	**10**	
Attendance	5	For 10 weeks
Coding problem sets	5	For around 10 practices
Examinations	**40**	
Quiz#1	5	Sections: 1-3
Mid-Term Exam	10	Sections: 1-5
Quiz#2	5	Sections: 6-8
Final Exam	20	**Comprehensive – ALL materials covered**
Homework	**40**	
Problem sets	5*8	For 8 assignments
Total	**100**	

Note that I reserve the right to change the distribution and add assignments as necessary, as a result, the final total grade may vary a bit as events of the quarter unfold and affect some assignments.

Your final letter grade in this course will be based on **curved credits** and follow the distribution below:

A	**93-100%**	**B**	**83-86%**	**C**	**73-76%**	**F**	**0-59%**
A-	**90-92%**	**B-**	**80-82%**	**C-**	**70-72%**		
B+	**87-89%**	**C+**	**77-79%**	**D**	**60-69%**		

Extra Credit(s) Opportunities

3 extra credits in total will be awarded to encourage lecture participation, extra effort on homework and course evaluation completion. These extra credits will be awarded as follows:

- Any student who asks or answers questions during the lectures will be awarded **0.1 extra credit**. Up to **1 extra credit** will be awarded throughout the course.
- Homework 5-8 will include some optional questions. Each time a student provides correct solutions to these questions will be awarded **0.3 extra credit**. Up to **1.5 extra credits** will be awarded in total.
- Students who complete the final course evaluation will be awarded **0.5 extra credit**.

Policies on Late Assignments

All writing/coding assignments will have specific due dates. Students are required to submit electronic versions of their solutions on Camino.

No late assignments will be accepted unless it is due to a problem that is beyond your control, e.g., medical or business related. Specifically excluded are personal and vacation plans. Contact the instructor as soon as you know that you have a conflict.

CLASSROOM COMPUTER USE

Cell phones, pagers, and similar electronic communication devices should be turned off during all classes. If you have an urgent phone call or text to respond to, please quietly exit the classroom and take care of it outside of our classroom or offline. You are welcome to use notebook computers or other devices to take notes as long as it does not hinder your ability to actively participate in class discussion.

ACADEMIC POLICY

Student responsibility: By enrolling in this class, a student is bound by the requirements stated in this syllabus and given by the instructor. Students must operate with integrity in their dealings with faculty and other students; engage the learning materials with appropriate attention and dedication; maintain their engagement when challenged by difficult learning activities; contribute to the learning of others; and perform to standards set by the faculty.

Classroom behavior: Students should behave properly during class time to facilitate the teaching and learning.

ACADEMIC INTEGRITY

The Academic Integrity pledge is an expression of the University's commitment to fostering an understanding of -- and commitment to -- a culture of integrity at Santa Clara University. The Academic Integrity pledge, which applies to all students, states:

I am committed to be a person of integrity. I pledge, as a member of the Santa Clara University community, to abide by and uphold the standards of academic integrity contained in the Student Conduct Code.

Students are expected to uphold the principles of this pledge for all work in this class. For more information about Santa Clara University's academic integrity pledge and resources about ensuring academic integrity in your work, see www.scu.edu/academic-integrity.

Work done for this course must adhere to the University's expectations outlined in the University Bulletin and the Student Handbook. The minimum penalty for cheating on an assignment is a failing grade. Cheating on exams will result in an F for the course. In addition, cheating will also be reported to the Office of Student Life according to the University Academic Integrity Protocol.

OFFICE OF ACCESSIBLE EDUCATION (FORMERLY DISABILITY RESOURCES)

If you have a documented disability for which accommodations may be required in this class, please contact the Office of Accessible Education (Benson 1, http://www.scu.edu/oae*, 408-554-4109) as soon as possible to discuss your needs and register for accommodations with the University. If you have already arranged accommodations through OAE, please discuss them with me during my office hours during the first two weeks of class.*

To ensure fairness and consistency, individual faculty members are required to receive verification from the Office of Accessible Education before providing accommodations. OAE will work with students and faculty to arrange proctored exams for students whose accommodations include double time for exams and/or assisted technology. Students with approved accommodations of time-and-a-half should talk with me as soon as possible.

OAE must be contacted in advance (at least two weeks of notice recommended) to schedule proctored examinations or to arrange other accommodations.

ACCOMMODATIONS FOR PREGNANT AND PARENTING STUDENTS

In alignment with Title IX of the Education Amendments of 1972, and with the California Education Code, Section 66281.7, Santa Clara University provides reasonable accommodations to students who are pregnant, have recently experienced childbirth, and/or have medically related needs related to childbirth. Pregnant and parenting students can often arrange accommodations by working directly with their instructors, supervisors, or departments. In addition, the Office of Accessible Education will provide reasonable accommodations for pregnancy-related impairments which impact a major life activity.

DISCRIMINATION AND SEXUAL MISCONDUCT (TITLE IX)

Santa Clara University upholds a zero-tolerance policy for discrimination, harassment and sexual misconduct. If you (or someone you know) have experienced discrimination or harassment, including sexual assault, domestic/dating violence, or stalking, I encourage you to tell someone promptly. For more information, please consult the University's Gender-Based Discrimination and Sexual Misconduct Policy at http://bit.ly/2ce1hBb *or contact the University's EEO and Title IX Coordinator, Belinda Guthrie, at 408-554-3043,* bguthrie@scu.edu. *Reports may be submitted online through* https://www.scu.edu/osl/report/ *or anonymously through Ethicspoint* https://www.scu.edu/hr/quick-links/ethicspoint/

IN-CLASS RECORDINGS

The Student Conduct Code (p. 13) prohibits students from "making a video recording, audio recording, or streaming audio/video of private, non-public conversations and/or meetings, inclusive of the classroom setting, without the knowledge and consent of all recorded parties," except in cases of approved disability accommodations. The Student Conduct Code also prohibits the "falsification or misuse, including non-authentic, altered, or fraudulent misuse, of University records, permits, documents, communication equipment, or identification cards and government-issued documents." Dissemination or sharing of any classroom recording without the permission of the instructor would be considered "misuse" and, therefore, prohibited. Violations of these policies may result in disciplinary action by the University. At the instructor's discretion, violations may also have an adverse effect on the student's grade.

ASSOCIATION WITH THE "VALUES IN SCIENCE AND TECHNOLOGY" PATHWAY

You can find information about Pathways on the Core Curriculum website https://www.scu.edu/provost/core/ *including specific Pathways, all courses associated with them, and the Reflection Essay prompt and rubric used to evaluate the final essay you will submit.* https://www.scu.edu/provost/core/pathways/pathway-reflection-essay-instructions/essay-submissions/

If you declare this Pathway, you may use a representative piece of work from this course as one of the Pathways materials you will upload via e-Campus during your junior or senior year. Therefore, we recommend that you keep electronic copies of your work using Dropbox or Google Docs, in addition to saving copies on your own computer or flash drives. This may ensure you will have a range of choices for retrieving your saved files when you analyze and assemble your Pathways materials in preparation to write the Pathways reflection essay.

TENTITIVE CLASS SCHEDULE

This schedule is subject to change with fair notice. Notification of any changes will be sent to all students.

Week #	Class Date/Day	Topics	Assignments and Class Activities
1	Apr 3/5	➢ Course Overview ➢ Syllabus Walkthrough ➢ zyBook Guide ➢ **Python Setup** ➢ **Section 1.1 – 1.6**: Introduction to Python	*Students should procure the required Textbook asap; Python installation* *Recommended reading & PA* ***Section 1.5 – 1.6***
2	Apr 8/10/12	➢ **Section 1.7 – 1.9**: Input Output	*Recommended reading & PA* ***Section 1.7 – 1.9***
3	Apr 15/17/19	➢ **Section 2.1 – 2.11**: Variables and Expressions	*Recommended reading & PA* ***Section 2.1 – 2.9*** *HW1 Assigned* ***HW1 Due Midnight Sunday***
4	Apr 22/24/26	➢ **Section 3.1 – 3.2**: Types ➢ **QUIZ #1**:	*Recommended reading & PA* ***Section 3.1 – 3.2*** *HW2 Assigned* ***QUIZ#1: Section 1-2 Friday*** ***HW2 Due Midnight Sunday***
5	Apr 29/ May 1/3	➢ **Section 3.4 – 3.7**: Types cont.	*Recommended reading & PA* ***Section 3.4 – 3.7*** *HW3 Assigned* ***HW3 Due Midnight Sunday***
6	May 6/8/10	➢ **Section 4.1 – 4.4**: Branching ➢ **Midterm Review** ➢ **MID-TERM**	*Recommended reading & PA* ***Section 4.1 – 4.4*** *HW4 Assigned* ***MID-TERM EXAM: Section 1-4 Friday*** ***HW4 Due Midnight Sunday***
7	May 13/15/17	➢ **Section 4.5 – 4.9**: Branching cont. ➢ **Section 5.1 – 5.4**: Loops	*Recommended reading & PA* ***Section 4.5 – 4.9*** *Recommended reading & PA* ***Section 5.1 – 5.4*** *HW5 Assigned* ***HW5 Due Midnight Sunday***
8	May 20/22/24	➢ **Section 5.5 – 5.11**: Loops cont. ➢ **Section 6.1 – 6.4**: Functions	*Recommended reading & PA* ***Section 5.5 – 5.11*** *Recommended reading & PA* ***Section 6.1 – 6.4*** *HW6 Assigned* ***HW6 Due Midnight Sunday***
9	May 29/31	➢ **Section 6.5 – 6.7**: Functions ➢ **QUIZ #2**	*Recommended reading & PA* ***Section 6.5 – 6.7*** *HW7 Assigned* ***QUIZ#2: Section 4-6 Friday*** ***HW7 Due Midnight Sunday***
10	Jun 3/5/7	➢ **Sections 7.1 – 7.4**: Files ➢ Course Review ➢ Final Exam study guide	*Recommended reading & PA* ***Section 7.1 – 7.4*** *HW8 Assigned* ***HW8 Due Midnight Sunday***
11	Jun 10 -14	➢ **FINAL** 9:15am Section: 9:10am, Jun 12 10:30am Section: 9:10am, Jun 10	***FINAL EXAM – Sections 1-7***